FNR 6462: Spring 2026

1. COURSE DESCRIPTION

Focuses on practical applications of AI in geospatial analysis for fields like resource management, agriculture, forestry, and marine sectors. Students will gain hands-on experience with Python and GeoAI tools for tasks such as data preprocessing, spatial data mining, image processing, and pattern recognition. The course also covers the use of Generative AI for spatial data analysis and big data processing, alongside ethical and societal considerations.

- Spring semester, 3 credits
- 100% online (synchronous with asynchronous lecture recordings)
- http://elearning.ufl.edu/

Course prerequisites: No formal prerequisites are required for this course. However, familiarity with Python programming (including syntax, editors, and package usage) and ArcGIS Pro is expected.

Instructors:

- **Dr. Hartwig Henry Hochmair**, Ft. Lauderdale Research & Education Center,
  phone: (954) 577-6317; e-mail: hhhochmair@ufl.edu
- **Dr. Amr Abd-Elrahman**, Gulf Coast Research and Education Center,
  phone: (813) 757-2283; e-mail: aamr@ufl.edu
- **Dr. Geraldine Klarenberg**, Gainesville campus,
  phone: (352) 273-0792; e-mail: gklarenberg@ufl.edu

Each of the three instructors will be responsible for teaching a distinct section of the course—covering the first, middle, and final thirds— as outlined in the course content table in Section 4.

Communication:

- Please use the Canvas e-mail system for fastest response

Office hours:

- Thursdays from 5-6 pm in Zoom (or by appointment)

Lectures:

- Fridays, 11:45am-2:30pm (period 5-7) via Zoom; links to recordings will be provided on the course website

Required reading materials:

- No course book is required

Recommended reading materials:

- Gao, S., Hu, Y., & Li, W. (2024). Handbook of Geospatial Artificial Intelligence. Boca Raton: CRC Press.
- Bolón-Canedo, V., L. Morán-Fernández, et al. (2024). "A review of green artificial intelligence: Towards a more sustainable future." Neurocomputing **599**: 128096.
- McGovern, A., I. Ebert-Uphoff, et al. (2022). "Why we need to focus on developing ethical, responsible, and trustworthy artificial intelligence approaches for environmental science." Environmental Data Science **1**: e6.
- Xu, J., W. Zhou, et al. (2021). "A Survey on Green Deep Learning." arXiv preprint.
- Mai, G., W. Huang, et al. (2024). "On the Opportunities and Challenges of Foundation Models for GeoAI (Vision Paper)." ACM Transactions on Spatial Algorithms and Systems **10**(2): 11.

- Li, W., S. T. Arundel, et al. (2024). "GeoAI for Science and the Science of GeoAI." Journal of Spatial Information Science **29**: 1-17.

Software requirements:

- Latest version of ArcGIS Pro. Download and installation instructions are provided during the course
- Latest version of Anaconda; download at https://www.anaconda.com/download
- Python 3.x, which comes with ArcGIS Pro installation and Anaconda
- Additional library packages will be introduced throughout the course

2. LEARNING OUTCOMES

The course objective is to provide students with the following competencies at the completion of the course:

1. Explain GeoAI's significance in natural resource management, agriculture, forestry, and marine ecosystems
2. Utilize Python tools for AI-based spatial data analysis in related subject areas
3. Assess strengths and limitations of different GeoAI techniques and data sources in natural resource management
4. Utilize GeoAI for simulation, image processing, spatial optimization, and big-data analysis in natural resource applications
5. Evaluate ethical and societal impacts of GeoAI in natural resource management

3. COURSE LOGISTICS

- For each grading item a due date and time is given, which is at the beginning of the next class (11:45am Fridays).
-
- This course is a distance education course taught as live lectures using the virtual classroom software Zoom. Links to lecture recordings will be posted on Canvas.
- Lecture materials can be downloaded from weekly modules on the Canvas website.
- This course does not offer extra credit tasks

The Canvas system should be used as the primary platform for written communication between students and the instructor. Questions and suggestions to the class can also be posted under the Discussions tab. Any short-term changes concerning lectures or other course components will be announced through Canvas. Feel free to contact the instructors with any questions.

Coursework requirements:

- Home assignments: Students will develop Python programming code and utilize analysis tools to apply AI methods for solving spatial problems in fields such as natural resources, Geomatics, forestry, fisheries, and other related areas. Assignments will be evaluated based on timeliness, accuracy, interpretation of numerical results, and technical proficiency. Written feedback will be provided by the instructor within one week. Detailed grading criteria for each assignment are available in the grading rubrics on the course website.

- Quizzes: Quizzes assess students' understanding of AI applications and data analysis principles in the context of natural resources and related disciplines. The quizzes consist of multiple-choice and matching questions, which will be automatically graded in Canvas upon submission.

- Topical discussions: Non-synchronous discussions will include one or more tasks focused on the application of AI in natural resources and related disciplines, designed to enhance students' critical thinking skills. To earn full points, students must submit at least one original post addressing the assigned task, along with at least

two responses to posts from their peers. Discussions will be graded within a week based on creativity, thoroughness, technical accuracy, and the quality and quantity of comments provided to fellow students.

- Home assignment discussions: These discussions are provided for each assignment, offering a platform for students to engage with open questions and challenges, thereby promoting interaction. To earn full points for this component, students must contribute at least one original post and one response to discussions for at least four different home assignments.
- Final group project: All students will collaborate in groups of 3 or 4 to develop an AI-related case study of their choice focused on natural resources or a related field. The goal of the case study is to collectively conduct an in-depth analysis of a geospatial problem through application of AI methods covered in the course. The process for developing the case study includes: (1) identifying the case study topic, (2) selecting data sources and analytical tools, (3) performing data analysis, (4) producing the final report, (5) project presentation. All members of the group will receive the same grade. Final group projects will be assessed based on creativity, thoroughness, technical accuracy, and quality of final report and presentation.

Technology Requirements:

- A computer or mobile device with high-speed internet connection and a headset and/or microphone and speakers to view lectures or join live sessions.
- ArcGIS Pro runs only on Microsoft operating systems. Students using a Mac computer, or other operating systems may consider using UF Apps (https://info.apps.ufl.edu/) instead which has ArcGIS Pro installed. Though this is an alternative, the instructors cannot provide any technical support when choosing that option.
- Anaconda and Python run on Windows and MacOS.
- For Zoom: A supported web browser on a supported operating system (Windows, Mac OS, Linux); and minimum bandwidth. More details can be found here.
- Canvas supports various browsers software, including Chrome, Firefox, Edge, and Safari, and JavaScript must be enabled. More information can be found here.

Using Zoom:

Live lectures and office hour (Q&A) meetings are conducted on Zoom. Sessions can be joined by clicking a link posted on Canvas.

Synchronous online sessions will be recorded. By sharing your video, screen, or audio during any synchronous online class sessions, you are consenting to being recorded for the benefit of students who cannot attend live as well as for class review during the current semester. If you have special circumstances or concerns about privacy, it is your responsibility to discuss it with your instructor.

Grades:

Item	***Percentage***
Home assignments (8 assignments @ 8.5% each)	68%
Quizzes (3 quizzes @ 3% each)	9%
Topical discussions (4 discussion @ 2.5% each)	10%
Final group project	8%
Home assignment discussions (4)	5%
Total	**100%**

Grading scale:

Grade	*Percentage*	*Grade*	*Percentage*
A	92.0-100.0	C+	78.0-79.9
A-	90.0-91.9	C	72.0-77.9
B+	88.0-89.9	C-	70.0-71.9
B	82.0-87.9	D	60.0-69.9
B-	80.0-81.9	E	0-59.9

For information on current UF policies for assigning grade points, see https://catalog.ufl.edu/UGRD/academic-regulations/grades-grading-policies/.

4. COURSE CONTENT

Week	Topic	Assessments
Week 1, Jan 16 (H)	Introduction to GeoAI in natural resources	D1, Q1
Week 2, Jan 23 (H)	Python review: Language basics, environments, notebooks, handling libraries and packages	H1
Week 3, Jan 30 (H)	Python libraries for GeoAI with natural resource examples	H2
Week 4, Feb 6 (H)	Generative AI and spatial analysis in natural resources	H3, D2
Week 5, Feb 13 (A)	Spatial optimization in natural resource management	H4
Week 6, Feb 20 (A)	GeoIA for spatial and temporal Geomatics applications	H5
Week 7, Feb 27 (A)	GeoAI based simulation in natural resource management	H6
Week 8, Mar 6 (A)	AI-based spatial image processing applications	H7, Q2
Mar 9 – Mar 13	*SPRING BREAK*	
Week 9, Mar 20 (K)	High-Performance computing and big data in GeoAI	H8
Week 10, Mar 27 (K)	GeoAI in agriculture and forestry	D3
Week 11, Apr 3 (K)	GeoAI in marine and fisheries management	P1
Week 12, Apr 10 (K)	Ethical and societal implications of GeoAI	Q3
Week 13, Apr 17 (K)	Course recap and future directions	D4

D: Discussion, Q: Quiz, H: Home assignment, P: Project
H…Hochmair, A…Abd-Elrahman, K...Klarenberg

5. POLICIES AND REQUIREMENTS

Late submissions and make-up requests:

It is the responsibility of the student to access on-line lectures, readings, quizzes, and assignments to maintain satisfactory progress in the course.

- A 10% penalty per day will be applied to late assignments. A late submission on the due date results also in a 10% deduction. Students can submit a late assignment as an attachment in an e-mail to the instructor.

- Assignments will not be accepted if handed in more than seven days after the due date.
- Students may request an extension to submitting a home assignment for a justified reason but need to do so before the submission deadline.
- Quizzes and discussions cannot be submitted past the deadline.

Computer or other hardware failures, except failure of the UF canvas system, will not excuse students for missing assignments. Any late submissions due to technical issues MUST be accompanied by the ticket number received from the Helpdesk when the problem was reported to them. The ticket number will document the time and date of the problem. Please e-mail your instructor within 24 hours of the technical difficulty if you wish to request consideration.

For computer, software compatibility, or access problems call the HELP DESK phone number—352-392-HELP = 352-392-4357.

Semester Evaluation Process:
Student assessment of instruction is an important part of efforts to improve teaching and learning.

At approximately the mid-point of the semester, the SFFGS will request anonymous feedback on student satisfaction on various aspects of this course. These surveys will be sent out through Canvas and are not required but encouraged. This is not the UF Faculty Evaluation!

At the end of the semester, students are expected to provide professional and respectful feedback on the quality of instruction in this course by completing course evaluations online via GatorEvals. Guidance on how to give feedback in a professional and respectful manner is available at https://gatorevals.aa.ufl.edu/students/. Students will be notified when the evaluation period opens, and can complete evaluations through the email they receive from GatorEvals, in their Canvas course menu under GatorEvals, or via https://ufl.bluera.com/ufl/. Summaries of course evaluation results are available to students at https://gatorevals.aa.ufl.edu/public-results/

Netiquette: Communication Courtesy Semester Evaluation Process:
All members of the class are expected to follow rules of common courtesy in all email messages, threaded discussions and chats, as laid out in the UF Netiquette Guide for Online Courses. Failure to do so may result in loss of participation points and/or referral to the Dean of Students' Office.

Diversity Statement:
This course encourages open and respectful communication among students, fostering a culture of inclusivity, understanding, and empathy. It values the unique backgrounds and experiences that each student brings to the classroom, including diverse racial and ethnic backgrounds, genders, sexual orientations, ages, abilities, religions, and socioeconomic statuses.

Academic Honesty Policy:
UF students are bound by The Honor Pledge which states "We, the members of the University of Florida community, pledge to hold ourselves and our peers to the highest standards of honor and integrity by abiding by the Honor Code. On all work submitted for credit by students at the University of Florida, the following pledge is either required or implied: "On my honor, I have neither given nor received unauthorized aid in doing this assignment." The Conduct Code specifies a number of behaviors that are in violation of this code and the possible sanctions. See the UF Conduct Code website for more information. If you have any questions or concerns, please consult with the instructor or TAs in this class. It is assumed that you will complete all work independently in each course unless the instructor provides explicit permission

for you to collaborate on course tasks (e.g. assignments, papers, quizzes, exams). It is your individual responsibility to know and comply with all university policies and procedures regarding academic integrity and the Student Honor Code. Violations of the Honor Code at the University of Florida will not be tolerated. Violations will be reported to the Dean of Students Office for consideration of disciplinary action. For more information regarding the Student Honor Code, please see: https://policy.ufl.edu/regulation/4-040/

University Policy on Accommodating Students with Disabilities:
Students with disabilities who experience learning barriers and would like to request academic accommodations should connect with the Disability Resource Center. See the "Get Started With the DRC" webpage on the Disability Resource Center site. It is important for students to share their accommodation letter with their instructor and discuss their access needs, as early as possible in the semester.

Privacy and Accessibility Policies:

- UF Privacy Policy
- Python Privacy Policy
- Anaconda Privacy Policy
- Google Privacy Policy
- ESRI Privacy Policy
- UF Accessibility Policy
- Google Accessibility Policy
- ESRI Accessibility Policy

Python, as a programming language, and Anaconda do not have a formal accessibility policy, but both the Python Software Foundation and Anaconda, Inc. promote inclusive practices within its operations, such as making its official documentation accessible and adhering to web standards.

Software use:
All faculty, staff and students at the university are required and expected to obey the laws and legal agreements governing software use. Failure to do so can lead to monetary damages and/or criminal penalties for the individual violator. Because such violations are also against university policies and rules, disciplinary action will be taken as appropriate.

Lecture recordings:
All live lectures and Q&A sessions will be recorded and made available via Canvas. Policies regarding student in-class recordings are detailed here http://aa.ufl.edu/policies/in-class-recording/.

6. CAMPUS RESOURCES
Students experiencing crises or personal problems that interfere with their general wellbeing are encouraged to utilize the university's counseling resources. The Counseling & Wellness Center provides confidential counseling services at no cost for currently enrolled students. Resources are available on campus for students having personal problems or lacking clear career or academic goals, which interfere with their academic performance.

Whole Gator App
The Whole Gator and website and app connects UF students with resources dedicated to supporting overall health and well-being. In addition to many of the resources below it also has strategies to practice self-care.
https://one.uf.edu/whole-gator/topics

Health and Wellness

- U Matter, We Care: If you or someone you know is in distress, please contact umatter@ufl.edu, 352-392-1575, or visit U Matter, We Care website to refer or report a concern and a team member will reach out to the student in distress.

- Counseling and Wellness Center: Visit the Counseling and Wellness Center website or call 352-392-1575 for information on crisis services as well as non- crisis services.

- Student Health Care Center: Call 352-392-1161 for 24/7 information to help you find the care you need, or visit the Student Health Care Center website.
- University Police Department: Visit UF Police Department website or call 352- 392-1111 (or 9-1-1 for emergencies).

- UF Health Shands Emergency Room / Trauma Center: For immediate medical care call 352-733-0111 or go to the emergency room at 1515 SW Archer Road, Gainesville, FL 32608; Visit the UF Health Emergency Room and Trauma Center website.

- GatorWell Health Promotion Services: For prevention services focused on optimal wellbeing, including Wellness Coaching for Academic Success, visit the GatorWell website or call 352-273- 4450.

- Student Success Initiative, http://studentsuccess.ufl.edu.

Academic Resources

- E-learning technical support: Contact the UF Computing Help Desk at 352-392- 4357 or via e-mail at helpdesk@ufl.edu.

- Career Connections Center: Reitz Union Suite 1300, 352-392- 1601. Career assistance and counseling services.

- Library Support: Various ways to receive assistance with respect to using the libraries or finding resources. Call 866-281-6309 or email ask@ufl.libanswers.com for more information.

- Teaching Center: 1317 Turlington Hall, 352-392-2010 or to make an appointment 352- 392-6420. General study skills and tutoring.

- Writing Studio: Daytime (9:30am-3:30pm): 2215 Turlington Hall, 352-846-1138 | Evening (5:00pm-7:00pm): 1545 W University Avenue (Library West, Rm. 339). Help brainstorming, formatting, and writing papers.

- Academic Complaints: Office of the Ombuds; Visit the Complaint Portal webpage for more information.

- Enrollment Management Complaints (Registrar, Financial Aid, Admissions): View the Student

Complaint Procedure webpage for more information.

Student Complaints:

- Residential Course: https://www.ombuds.ufl.edu/complaint-portal/
- Online Course: https://pfs.tnt.aa.ufl.edu/state-authorization-status/#student-complaint